

USB4 1.0 ENGINEERING CHANGE NOTICE FORM

Title: Shared Buffers Credit Sync on Path Teardown Applied to: USB4 Specification Version 1.0

Brief description of the functional changes:

On a Path teardown, if ESE is set to 1b, then a router is mandated to send a Shared Buffers Credit Sync Packet. The change is to mandate this behavior only if there is at least one additional enabled Path that uses Egress Shared Buffering.

Benefits as a result of the changes:

This change is to align to the requirement that a Shared Buffers Credit Sync Packet is mandated to be sent if there is at least one enabled Path that uses shared buffers.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

None

An analysis of the hardware implications:

None

An analysis of the software implications:

None

An analysis of the compliance testing implications:

None

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Actual Change

(a). Section 5.4.1 Egress Adapter (Path Teardown)

5.4.1 Egress Adapter

For the Egress Adapter of the Path being torn down, a Router shall perform the following steps in the order listed:

1. If the *ESE Flag* field in the Path Configuration Space is set to 1b for the Path and besides the Path being torn down there is additional enabled Path with its *ESE Flag* set to 1b, then the Router shall send a Shared Buffers Credit Sync Packet to the Link Partner.
2. While the *Valid* bit in the Path Configuration Space is 0b:
 - The Router shall not send any Path Credit Sync Packets for the Path.
 - The Router shall ignore any Path credit updates for the Path received on the Egress Adapter.
3. The Router may transmit packets on the Path when the *Pending Packets* bit in the Path Configuration Space is set to 1b and for tTeardown after the Router sets the *Pending Packets* bit to 0b.
4. After tTeardown time, the Router shall discard any remaining packets for the Path.
5. The Router shall block the transmission of any packets on the Path after tTeardown has elapsed since it set the *Pending Packets* bit to 0b and until the *Valid* bit in the Path Configuration Space is set again to 1b.